

Subject : DS-505: Big Data Handling and Management for Machine Learning Applications
Course : Bachelors of Science in Data Science and Analytics Semester : Semester-5
Model Paper 2

Q.1 A	Answer the following in short (Any 4)	04 Marks
1	Which characteristic of Big Data refers to the trustworthiness, noise level, and quality of data? a) Velocity b) Veracity c) Variety d) Volume	
2	In PySpark, which method is used to display the structural tree hierarchy of columns and their data types? a) df.show() b) df.printSchema() c) df.describe() d) df.columns()	
3	Which strategy is most suitable for imputing missing values in a numerical column that has extreme outliers? a) Mean Imputation b) Median Imputation c) Zero Imputation d) Drop the entire dataset	
4	Which neural network architecture forms the core foundation of modern Large Language Models? a) CNN b) RNN c) Self-Attention Transformer d) MLP	
5	What does test_size=0.20 signify in Scikit-Learn's train_test_split()? a) 20 samples reserved for test b) 20% of dataset allocated for testing c) 80% of dataset discarded d) Random seed set to 20	
Q.1 B	Answer the following in short (Any 4)	04 Marks
1	Define Big Data according to Doug Laney's 3V taxonomy.	
2	What does .master("local[*]") signify in a PySpark SparkSession?	
3	Why do we use SimpleImputer in Scikit-Learn?	
4	What is Tokenization in Natural Language Processing (NLP)?	
5	What is an Out-Of-Memory (OOM) error in Python?	
Q.2 A	Write a short note on the following (any one)	05 Marks
1	Explain the 5 Vs of Big Data in detail with suitable real-world industrial examples.	
2	Explain the advantages of Apache Spark over traditional Hadoop MapReduce. Why is Apache Spark significantly faster for data processing?	
B	Compare CSV, JSON, and Apache Parquet file formats. Which format is best suited for Big Data applications and why?	04 Marks
Q.3	Write a short note on the following (any Two)	08 Marks
1	What is Feature Engineering? Explain the difference between Label Encoding and One-Hot Encoding with simple examples.	
2	Explain the process of Text Summarization using Large Language Models. How does it benefit modern organizations in handling large volumes of text?	
3	Differentiate between Data Normalization (Min-Max Scaling) and Data Standardization (Z-Score). Write their simple formulas and explain when each should be used.	